System Architecture Diagram & Data Flow

Project: Smart Irrigation Alert Dashboard
Student: Chetan Katkar — **Roll No:** 23102A0024 — **Class:** BE CMPN-A

1. High-Level System Architecture

```
flowchart TB
    subgraph ClientTier["Client / Presentation Tier"]
        Browser["Modern Web Browser<br>(HTML5, CSS3, Vanilla JS, Bootstrap Icons)"]
    end

    subgraph AppTier["Application Tier (Spring Boot / Java 21)"]
        Controller["SensorWebController & SensorApiController<br>(Spring MVC REST Controllers)"]
        Service["SensorAlertService<br>(Business Logic & Threshold Engine)"]
        AlertEngine["Alert Exception Engine<br>(Moisture < 30% Trigger)"]
    end

    subgraph DataTier["Data Tier"]
        Repo["SensorRepository (Spring Data JPA)"]
        H2[("H2 In-Memory / File Database")]
    end

    subgraph OpsTier["DevOps & Infrastructure Tier"]
        Jenkins["Jenkins Pipeline CI/CD"]
        Selenium["Selenium WebDriver Test Suite"]
        Docker["Docker Engine & Container"]
        Ansible["Ansible Configuration Playbook"]
    end

    Browser <-->|HTTP GET/POST /api/sensors JSON & HTML| Controller
    Controller <--> Service
    Service <--> AlertEngine
    Service <--> Repo
    Repo <--> H2

    OpsTier -.->|Build, Test & Package| AppTier
    OpsTier -.->|Automated UI Validation| Browser
```

2. Data Model & Entity Specification

Entity: SensorZone

```
public class SensorZone {
    private Long id;              // Unique primary key (Auto-increment)
    private String zoneName;      // e.g., "Zone A - North Vineyard"
    private String cropType;      // e.g., "Wheat", "Tomato", "Cotton"
    private double moistureLevel; // Current soil moisture (0.0 to 100.0%)
    private double temperature;   // Ambient temperature in Celsius
    private String status;        // "CRITICAL" (<30%), "OPTIMAL" (30-60%), "SATURATED" (>60%)
```

```
    private LocalDateTime lastUpdated; // Timestamp of reading
}
```